Homework 7: Projections; Regression using Linear Algebra

EECS 245, Spring 2026 at the University of Michigan

due Thursday, June 4th, 2026 at 11:59PM Ann Arbor Time (no slip days allowed!)

Write your solutions to the following problems either by writing them on a piece of paper or on a tablet and scanning your answers as a PDF. Note that you are not allowed to use LaTeX, Google Docs, or any other digital document creation software to type your answers. Homeworks are due to Gradescope by 11:59PM on the due date. See the **syllabus** for details on the slip day policy.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should always explain and justify your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Before proceeding, make sure you're familiar with the **collaboration policy**.

Total Points: 10 + 6 + 18 + 27 + 13 + 8 + 14 = 96

Problem 1: Homework 6 Solutions Review (10 pts)

Review the solutions to Homework 6. Pick **two problem parts** (for example, Problem 2a and Problem 5b) from Homework 6 in which your solutions have the most room for improvement, i.e., where they have unsound reasoning, could be significantly more efficient or clearer, etc. **Include a screenshot of your solution to each problem part**, and in a few sentences, explain what was deficient and how it could be fixed.

Alternatively, if you think one of your solutions is significantly better than the posted one, copy it here and explain why you think it is better. If you didn't do Homework 6, choose two problem parts from it that look challenging to you, and in a few sentences, explain the key ideas behind their solutions in your own words.

Problem 2: Anonymous Feedback (6 pts)

We'd like to get your feedback on how the course has been going so far, now that we're past the halfway point and Midterm 2 is fast approaching.

You can find the survey **at this link**, which you should complete **after you've finished the rest of Homework 7**. Unlike the Homework 3 survey, **it is anonymous**, so feel free to provide candid feedback.

In order to earn the 6 points for Homework 7, Problem 2, include a screenshot of the confirmation message you see after submitting the form. (We consider it an honor code violation to include a screenshot if you didn't actually submit the form!)

Thank you for your feedback once again!

Problem 3: The Complete Solution (18 pts)

Before beginning this problem, make sure you've read both **Chapter 6.3** and **Chapter 6.4**.

a) (4 pts) Consider the matrix A and vector $\vec{b}$ defined below.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 2 & -1 \\ 1 & -1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$$

Find the vector $\vec{x}^*$ that minimizes $\|\vec{b} - A\vec{x}\|^2$. Show your work, but feel free to use `numpy` to compute some of the matrix operations; **here** is a video that walks through how to do this. (We're intentionally using different variables here than in the notes to have you think about the problem in general terms.)

For the remainder of the problem, we will use the same vector $\vec{b}$, but instead use the following matrix A.

$$A = \begin{bmatrix} 1 & 0 & 1 & -4 & 4 \\ 0 & 2 & 1 & -5 & 3 \\ 2 & -6 & -1 & 7 & -1 \\ 1 & -4 & -1 & 6 & -2 \end{bmatrix}$$

Note that this new matrix A has **a rank of 2**.

b) (2 pts) Now, find **one** vector $\vec{x}^*$ that minimizes $\|\vec{b} - A\vec{x}\|^2$ for this new matrix A. Again, show your work. If you've read **Chapter 6.4** closely, this should not require much calculation.

c) (4 pts) Find a basis for $\text{nullsp}(A)$. *Hint: Try and do so efficiently, since this is the type of problem we'll see on Midterm 2.*

d) (2 pts) Show that if $\vec{x}'$ satisfies the normal equation, $A^TA\vec{x}' = A^T\vec{b}$, and $\vec{x}_0 \in \text{nullsp}(A)$, then $\vec{x}' + \vec{x}_0$ also satisfies the normal equation. *Hint: This is two-line solution; we're mostly asking it so that you interalize **what** this means and why it's true.*

e) (3 pts) Describe, using set notation, the complete set of vectors $\vec{x}^*$ that minimize $\|\vec{b} - A\vec{x}\|^2$. Is this set a subspace?

f) (3 pts) There are infinitely many vectors $\vec{x}^*$ that minimize $\|\vec{b} - A\vec{x}\|^2$. If we try and use code to find a solution, it can't return all of them — it'll pick a particular one.
In Python, use `np.linalg.lstsq` to find a vector $\vec{x}^*$ that minimizes $\|\vec{b} - A\vec{x}\|^2$. Include a screenshot of your code and the vector $\vec{x}^*$ it returns, and in your PDF, write out the coefficients of $\vec{x}^*$ as a vector (in addition to the screenshot). Then, provide an educated guess of **why** you think it picked the $\vec{x}^*$ that it did.

Problem 4: Orthogonalization (27 pts)

Before starting, refer to Chapter 6.5, written just for this problem. It won't be possible to do this problem without referencing it.

In parts **a)** through **d)**, we'll refer to the vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

a) (6 pts) By hand, apply the Gram-Schmidt process to the vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ to find an orthonormal set of vectors $\vec{q}_1, \vec{q}_2, \vec{q}_3$. Show your work; you cannot use `numpy` for this.

Then, create the matrix $Q = \begin{bmatrix} | & | & | \\ \vec{q}_1 & \vec{q}_2 & \vec{q}_3 \\ | & | & | \end{bmatrix}$ and confirm that $Q^TQ = I$, but that $QQ^T \neq I$.

b) (3 pts) Suppose $\vec{v}_4 = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 3 \end{bmatrix}$. If you were to apply the Gram-Schmidt process to the vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4$, what would the vector $\vec{Q}_4$ be? Why?

c) (4 pts) **[Updated version]** Consider $\vec{y} = \begin{bmatrix} 3 \\ 1 \\ 2 \\ 4 \end{bmatrix}$. $\vec{y}$ **is** in $\text{span}(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}) = \text{span}(\{\vec{q}_1, \vec{q}_2, \vec{q}_3\})$.

Find scalars a, b, and c such that $a\vec{q}_1 + b\vec{q}_2 + c\vec{q}_3 = \vec{y}$, **without** solving a system of 3 equations and 3 unknowns. Instead, use the fact that $\vec{q}_1, \vec{q}_2, \vec{q}_3$ are orthonormal. *Hint: There's a relevant problem from Lab 5.*

d) (4 pts) Consider $\vec{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. Unlike in **c)**, $\vec{y}$ **is not** in $\text{span}(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\})$.

Find the vector in $\text{span}(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\})$ that is closest to $\vec{y}$. **Do not** stack the $\vec{v}_i$'s into a matrix X and then use $X(X^TX)^{-1}X^T\vec{y}$. Instead, use the fact that $\vec{q}_1, \vec{q}_2, \vec{q}_3$ are orthonormal and have the same span as $\vec{v}_1, \vec{v}_2, \vec{v}_3$. How does this simplify the problem?

e) (5 pts) Open the **the supplemental Jupyter Notebook** we've created for Homework 7, which can either be found here in the course GitHub repository, or here on DataHub.

There, you're asked to implement the function `orthogonalize`, which takes in an $n \times d$ matrix V whose columns are linearly independent, and returns a matrix Q whose columns are orthonormal and have the same span as V. This problem is **not autograded**. Rather, in your submission to this part, include a screenshot of your implementation and sample output in your PDF for Homework 7.

f) (5 pts) A QR decomposition of a matrix A is a factorization of the form

$$A = QR$$

where Q is an $n \times d$ matrix with orthonormal columns and R is an $d \times d$ **upper triangular** matrix (a matrix that has 0s below the diagonal).

For example, if $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, a QR decomposition of A is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ -1/\sqrt{3} & 2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} \sqrt{3} & 2\sqrt{3}/3 & 2\sqrt{3}/3 \\ 0 & \sqrt{6}/3 & 5\sqrt{6}/6 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}}_{R}$$

Finding the Q in a QR decomposition is straightforward: apply Gram-Schmidt to the columns of A, assuming A's columns are linearly independent. The question is how to find R.

(i) In the supplemental Jupyter Notebook, we've defined an arbitrary matrix A and call your `orthogonalize` function on it, and give you hints as to how to find R. Using the experimentation there, and what you know about Q, **explain how to find** R**.**

(ii) Find a QR decomposition of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Note that the columns of A are made up of the same three vectors you worked with in parts **a)** through **d)** of this problem.

(iii) Given **a** QR decomposition of A, explain how to find **another** QR decomposition of A with a (slightly) different Q and/or R.

Problem 5: Same, but Different (13 pts)

In **Chapter 2.4**, we were introduced to one of many formulas for the optimal slope, w_1^*, and optimal intercept, w_0^*, for the simple linear regression model $h(x_i) = w_0 + w_1 x_i$ when using squared loss:

$$w_1^* = r\frac{\sigma_y}{\sigma_x} \qquad w_0^* = \bar{y} - w_1^* \bar{x}$$

The end goal of Chapters 3 through 6 has been to give us the tools to revisit the simple linear regression model in terms of linear algebra, so that we can extend our model to allow for multiple input variables. As we see in **Chapter 7.1**, the solution is to define the $n \times 2$ "**design matrix**" X and observation vector $\vec{y} \in \mathbb{R}^n$ as follows:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Then, the vector containing the optimal model parameters is

$$\vec{w}^* = (X^TX)^{-1}X^T\vec{y} = \begin{bmatrix} w_0^* \\ w_1^* \end{bmatrix}$$

It's not immediately obvious why the components of $\vec{w}^*$ should have anything to do with the correlation, means of x and y, and standard deviations of x and y. In this problem, we will prove that both of these formulations are equivalent, for any dataset (x_1, y_1), (x_2, y_2), ..., (x_n, y_n).

a) (5 pts) Express the matrix $(X^TX)^{-1}$ using constants and/or summations involving x_i and/or y_i.

b) (3 pts) Prove that

$$(X^TX)^{-1} = \frac{1}{n\sigma_x^2}\begin{bmatrix} \sigma_x^2 + \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix}$$

Hint: Start by proving $\sum_{i=1}^n x_i^2 = n\sigma_x^2 + n\bar{x}^2$.

c) (5 pts) Finally, prove that

$$(X^TX)^{-1}X^T\vec{y} = \begin{bmatrix} \bar{y} - r\frac{\sigma_y}{\sigma_x}\bar{x} \\ r\frac{\sigma_y}{\sigma_x} \end{bmatrix}$$

Hint: Start by proving that $\sum_{i=1}^n x_i y_i = nr\sigma_x\sigma_y + n\bar{x}\bar{y}$.

Note that the second component of the vector above is $w_1^* = r\frac{\sigma_y}{\sigma_x}$ and the first component of the vector above is $w_0^* = \bar{y} - r\frac{\sigma_y}{\sigma_x}\bar{x} = \bar{y} - w_1^*\bar{x}$, as we first saw in Chapter 2.4! I think this is beautiful.

Problem 6: Putting it into Practice (8 pts)

This problem asks you to apply the concepts in **Chapter 7.2**, and follows the last problem.

Suppose we'd like to fit a hypothesis function of the form $h(x_i) = w_0 + w_1 x_i^2$. Notice the squared term; this is **not** a simple linear regression line.

To do so, we'll find the optimal parameter vector $\vec{w}^*$ that satisfies the normal equations. The first 5 rows of our dataset are as follows, though note that our dataset has n rows in total.

x_i	y_i
2	4
-1	2
3	5
-7	3
3	-7
$\vdots$	$\vdots$

Suppose $x_1, x_2, ..., x_n$ have a mean of $\bar{x} = 5$ and a variance of $\sigma_x^2 = 8$.

a) (2 pts) Write the first five rows of the design matrix, X.

b) (3 pts) Suppose that after solving the normal equations, we find $\vec{w}^* = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$. Find the augmented feature vector $\text{Aug}(\vec{x}_3)$ and squared loss for row 3 of the dataset.

c) (3 pts) Let $X_{\text{tri}} = 3X$, where X is the full design matrix for our dataset with n rows. Determine the bottom-left value in the matrix $X_{\text{tri}}^T X_{\text{tri}}$, i.e. the value in the second row and first column. Your answer should be an expression involving n. *Hint: You can use any of the hints or results from Problem 2 without needing to re-prove them.*

Problem 7: Billy the Waiter (14 pts)

This problem involves writing code and submitting it to the Gradescope autograder. The goal of this problem is to give you a taste of how linear algebra can be used to implement linear regression in code, and show you how to build models that involve multiple features (including categorical variables).

Open the **the supplemental Jupyter Notebook** we've created for Homework 7, which can either be found **here** in the course GitHub repository, or **here** on DataHub.

This problem is entirely autograded; to receive credit for Problem 7 of this homework, you'll need to submit your completed notebook to the autograder on Gradescope. Your submission time for Homework 7 is the **latter** of your PDF and code submission times.